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SQL Functions

SQL Functions are built-in programs that are used to perform different operations on the database.

There are **two types** of functions in SQL: (**Aggregate Functions** - **Scalar Functions**)

SQL Aggregate Functions

operate on a data group and return a singular output. They are mostly used with the **GROUP BY** clause to summarize data.

Some common Aggregate functions with Syntax and description are shown in the table below.

<i>Aggregate Function</i>	<i>Description</i>	<i>Syntax</i>
<i>AVG()</i>	Calculates the average value	SELECT AVG(column_name) FROM table_name;
<i>COUNT()</i>	Counts the number of rows	SELECT COUNT(column_name) FROM table_name
<i>FIRST()</i>	Returns the first value in an ordered set of values	SELECT FIRST(column_name) FROM table_name;
<i>LAST()</i>	Returns the last value in an ordered set of values	SELECT LAST(column_name) FROM table_name;
<i>MAX()</i>	Retrieves the maximum value from a column	SELECT MAX(column_name) FROM table_name;
<i>MIN()</i>	Retrieves the minimum value from a column	SELECT MIN(column_name) FROM table_name;

SQL Scalar functions

SQL Scalar Functions are built-in functions that operate on a single value and return a single value.

Scalar functions in SQL helps in efficient data manipulation and simplification of complex calculations in SQL queries.

<i>Scalar function</i>	<i>Description</i>	<i>Syntax</i>
UCASE()	Converts a string to uppercase	SELECT UCASE(column_name) FROM table_name;
LCASE()	Converts a string to lowercase	SELECT LCASE(column_name) FROM table_name;
MID()	Extracts a substring from a string	SELECT MID(column_name, start, length) FROM table_name;
LEN()	Returns the length of a string	SELECT LEN(column_name) FROM table_name;
ROUND()	Rounds a number to a specified number of decimals	SELECT ROUND(column_name, specified number of decimals) FROM table_name;
NOW()	Returns the current date and time	SELECT NOW();
FORMAT()	Formats a value with the specified format	SELECT FORMAT(column_name, format) FROM table_name;

In SQL, Both built-in and user-defined functions can return either a **single (scalar) value** or an entire **table (result set)**, depending on how they are defined and used.

Functions that Return a Single Value (Scalar Functions)

Most common built-in SQL functions return a single, scalar value. These can be used in most places where an expression is valid

(e.g., in SELECT, WHERE, or HAVING clauses).

User-defined scalar functions can also be created to encapsulate complex calculations and return a single value.

Functions that Return a Table (Table-Valued Functions)

Both built-in system functions and user-defined functions can be designed to return a result set in the form of a table.

Built-in examples:

System functions like OPENJSON (in SQL Server) return a table structure which can then be queried like any other table.

Source : <https://www.geeksforgeeks.org>